

Markov Chains

Problem 1

The population of Replicant skin cells has a $4/5$ chance of tripling or $1/5$ chance of going extinct. What is the probability Replicant skin cells do not go extinct, given the Replicant population began with only one cell?

Solution 1

1. Let p be the probability of extinction
2. Construct the CKE at one cell $p = \frac{4}{5} \cdot p \cdot p \cdot p + \frac{1}{5} \cdot 1$, this is the case as $1/5$ chance of going extinct but if replicates, recursively all the triplicates must go extinct
3. Solve this equation $4p^3 - 5p + 1 = 0$ by first inspecting $p - 1$ is a root to factor into $(p - 1)(4p^2 + 4p - 1)$ via inspection then using quadratic formula $p = 1, 1/2(-1 + \sqrt{2}), 1/2(-1 - \sqrt{2})$
4. Since probability p is $0 \leq p \leq 1$ and intuitively we know Replicant cells have a chance of living so $p = 1/2(-1 + \sqrt{2})$

Problem 2

Fordo and Frodo are betting opposing wagers of \$1 on unbiased coin tosses, playing until one goes bankrupt. Fordo has \$99 and Frodo has \$66, what are Fordo's chances of losing?

Solution 2

This is a specific scenario where a Markov process models Gambler's ruin. Use the general $a/(a + b)$ formula to find Fordo's chances of losing as $\frac{99}{99+66} = \frac{3}{3+2} = \frac{3}{5}$.

Problem 3

Let's play a game with 2 coins. Coin 1 lands on heads 60% of the time, while coin 2 lands on heads 50% of the time. A coin is continually flipped until it lands on tails, then we switch to the other coin and repeat the coin flips until we have to switch again. If we start the game with coin 1, what is the probability that coin 2 is used on the 5th flip?

Solution 3

We have the transition matrix P given by

$$P = \begin{pmatrix} 0.6 & 0.4 \\ 0.5 & 0.5 \end{pmatrix},$$

so we can compute that

$$P^4 \approx \begin{pmatrix} 0.5556 & 0.4444 \\ 0.5555 & 0.4445 \end{pmatrix},$$

so $P_{12}^{(4)} \approx 0.4444$, given that we start on coin 1.

Problem 4

Suppose an organisation has n employees, and that each employee has one of three possible jobs, which changes every day according to a Markov chain with the following transition matrix:

$$\begin{pmatrix} 0.7 & 0.2 & 0.1 \\ 0.2 & 0.6 & 0.2 \\ 0.1 & 0.4 & 0.5 \end{pmatrix}.$$

Over the long-term, what percentage of employees will be in each job?

Solution 4

The solution is the limiting probabilities of the Markov chain, which exist since the necessary conditions are met. So, we just need to solve

$$\begin{aligned} \pi &= \pi P, \\ \sum_i \pi_i &= 1, \end{aligned}$$

which, if we rearrange all the equations, equates to row-reducing the matrix

$$\begin{pmatrix} 1 & 1 & 1 & 1 \\ 0.3 & -0.2 & -0.1 & 0 \\ 0.2 & -0.4 & 0.4 & 0 \end{pmatrix}.$$

In doing so, we obtain that

$$\pi = \left(\frac{6}{17} \quad \frac{7}{17} \quad \frac{4}{17} \right),$$

so for every 17 employees, 6 will be in job 1, 7 will be in job 2, 4 will be in job 3.

Problem 5

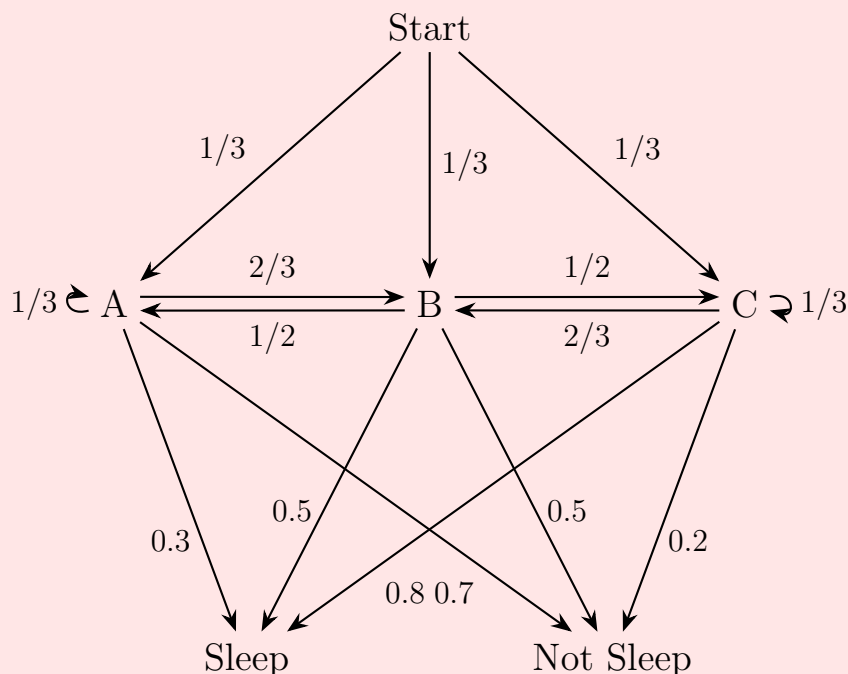
Scientists are measuring the propensity of the Janx species of animals on the planet Anaxes to sleep given the conditions A : sunny, B : sunset/sunrise, C : dark. There is an equal chance of the test beginning with each state. A to B is as likely as C to B , but both are twice as likely as A to A , C to C but A to C is impossible. It is impossible for C to C , C to A , and A to C but B to C , B to A are equally likely on this planet. Now given it is A , there is only a 30% change of falling asleep, 50% for B , and 80% for C .

1. Draw the Hidden Markov Model Diagram
2. What is the overall chance of falling asleep.

Solution 5

1.

Note one thing this is not the complete HMM for sleeping, there are other factors influencing asleep and we have not described how to transition from sleep to not sleep.



2.

To DOUBLE CHECK

$P(\text{sleep}) = 0.3P(A) + 0.5P(B) + 0.8P(C)$. Now we need to compute the probabilities of going to A from start. This will involve CKE solving. We solve the system

$$\begin{aligned} 2/3\pi_A &= 1/2\pi_B \\ \pi_A + \pi_B + \pi_C &= 1 \\ \pi_C &= \pi_A \quad (\text{symmetry}) \end{aligned}$$

Upon solving the first $\pi_B = 4/3\pi_A$, and applying this along with the third equation to the second equation above, $10/3\pi_A = 1$ so $P(A) = \pi_A = 3/10$. Consequently, $P(C) = \pi_C = 3/10$, and $P(B) = \pi_B = 4/3\pi_A = 4/10$, which means $P(\text{sleep}) = 0.3 \cdot 0.3 + 0.5 \cdot 0.4 + 0.8 \cdot 0.3 = 0.53$.

Problem 6

**Tintin and Justin play a game of chance where Tintin out of an infinite pile of a_1, a_2, a_3, a_4 cent coins and Justin out of b_1, b_2, b_3, b_4 cent coins randomly chose a denomination each round to add to their pile and whoever first reaches an integer amount wins the match and with it right to buy an All or Nothing Option at \$8 for the next match that pays off \$10 if their pile for that match reaches a winning sum under \$2.

1. Intuitively what is the EV of Tintin and Justin's pile?

2. *If Justin has a greater EV in his pile, how are his odds of profiting compared to Tintin from the All or Nothing Option, assuming he wins the previous match and buys the option?
3. Given Justin has 5, 10, 15, 25 cent coins and Tintin has 1, 2, 5, 10 cent coins, approximate Justin's Expected Profit for the second match, assuming he won the first won and bought the option.
4. What should be a fair price for this option given previous approximations?

Solution 6

1.

Intuitively $EV_{\text{Tintin}} = 100/4 \sum_{i=1}^4 a_i$, $EV_{\text{Justin}} = 100/4 \sum_{i=1}^4 b_i$.

2.

Intuitively $P(\text{Justin winning sum} < \$2) < P(\text{Justin Sum} = \$1) \simeq 1/EV_{\text{Justin}}$. Since $EV_{\text{Justin}} > EV_{\text{Tintin}}$, Justin's approximate chances of hitting a \$1 sum are less than Tintin's, so qualitatively he has bad odds compared to Tintin which means conditioning on Tintin if he had that option, Justin is more likely to receive a worthless option and make a \$8 loss.

3.

Simple Computation, Justin has $EV = 1/4 \cdot (55) = 13.75$ in his pile so his probability of winning sum reaching a \$1 is approximately $1/13.75$ so his expected gains as $1/13.75 \cdot 10 + 0$ meaning his expected profit is $10/13.75 - 8 \simeq -7.27$.

4.

Fair price for Justin when expected cost equals expected gain so $10/13.75 \simeq \$0.73$. Fair price for Tintin is similarly $10/(EV_{\text{Tintin}}) = 10/(18/4) \simeq \2.22 . Assuming they are the only two players in a market, a good estimate would be the mid price, assuming both bid and ask at their own fair price, the market fair price is approximately $(2.22 + 0.73)/2 \simeq 1.98$.